

Toxic Comment Classification

Comparative Study - Project 2

Modern NLP Approaches

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Compare 4 Modern NLP Approaches

- Bag-of-Words (BoW) with TF-IDF
- Word Embeddings (GloVe)
- Transformers (toxic-bert)
- Prompt Engineering

Project 1 Recap: Baseline Model

Logistic Regression + TF-IDF

Strengths

- **Fast:** 126s processing time
- **Excellent performance:** $F1=0.96$, $AUC=0.9757$
- **Interpretable:** TF-IDF weights reveal toxic keywords
- **Low cost:** No GPU required

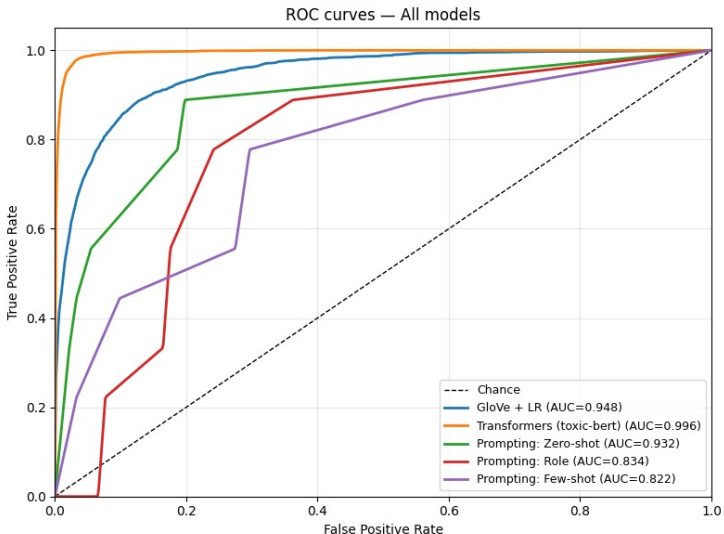
Weaknesses

- **No context:** "You're not stupid" vs "You're stupid"
- **Sparse vectors:** High dimensionality
- **OOV problem:** Unknown words ignored
- **No semantics:** Can't generalize to synonyms

Performance Comparison

Model	Precision	Recall	F1-Score	AUC	Time (min)
LogisticRegression	0.96	0.96	0.96	0.9757	2
GloVe + LR	0.93	0.89	0.90	0.9478	3
Transformers (toxic-bert)	0.98	0.98	0.98	0.9955	46
Zero-shot	0.92	0.45	0.54	0.9316	1
Role prompting	0.92	0.12	0.07	0.8339	1
Few-shot	0.88	0.73	0.78	0.8217	3

ROC Curves Analysis



Key Insight

Computational Cost & Deployment

Processing Time Comparison

- **Fastest:** Zero-shot (59.71s) and Role prompting (68.01s)
- **Intermediate:** GloVe + LR (186.76s) and Few-shot (199.62s)
- **Slowest:** Transformers (2777s, 46 minutes)

Computational Cost Analysis

Transformers

- Excellent performance but prohibitive cost
- Difficulty for real-time processing
- **Solutions:** distilled models, batch processing

Embeddings

- Reasonable compromise between speed and performance

Prompting

- Fast but insufficient performance
- Potentially high API costs

Bias: "She is a woman" Problem

Scenario

Neutral sentence "**She is a woman**" flagged as toxic

Why This Happens?

- **Biased corpus:** Over-representation of sexist comments
- **Skewed co-occurrences:** "woman" statistically linked to toxic contexts
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- **Over-generalization:** Demographic keywords mistaken for toxicity
- **Embedding bias:** GloVe/FastText inherit historical web biases

Transparency & Explainability

Scenario

User appeals moderation decision: **"You're being ridiculous"**

Explainable Models

- **GloVe + LR**: Feature weights interpretable
- Can identify influential words
- Clear decision rationale

Black-box Models

- **Transformers**: Complex attention mechanisms
- Requires SHAP/LIME for interpretation
- Hard to explain specific decisions

What Happens with External LLM APIs?

- **Data exposure:** Comments sent to third-party servers
- **Logs retention:** May be stored temporarily (30+ days)
- **Cross-border transfer:** Data leaves EU jurisdiction
- **Trust-dependent:** Zero-retention policies not verifiable

Risks for Wikipedia

Critical Risks

- **Data breach:** Leaking contributor comments & IPs
- **GDPR violations:** Non-compliant data transfers
- **Community trust loss:** Privacy expectations violated
- **Adversarial probing:** Attackers learn evasion tactics

Solutions

- On-premise model deployment
- Data anonymization techniques
- Strict contractual agreements

Conclusion

Key Takeaways

- Transformers offer best performance but require significant resources
- Final choice depends on latency, budget, and confidentiality requirements
- Critical to consider ethical aspects: bias, transparency, privacy
- No one-size-fits-all solution - context matters

Thank you for your attention!